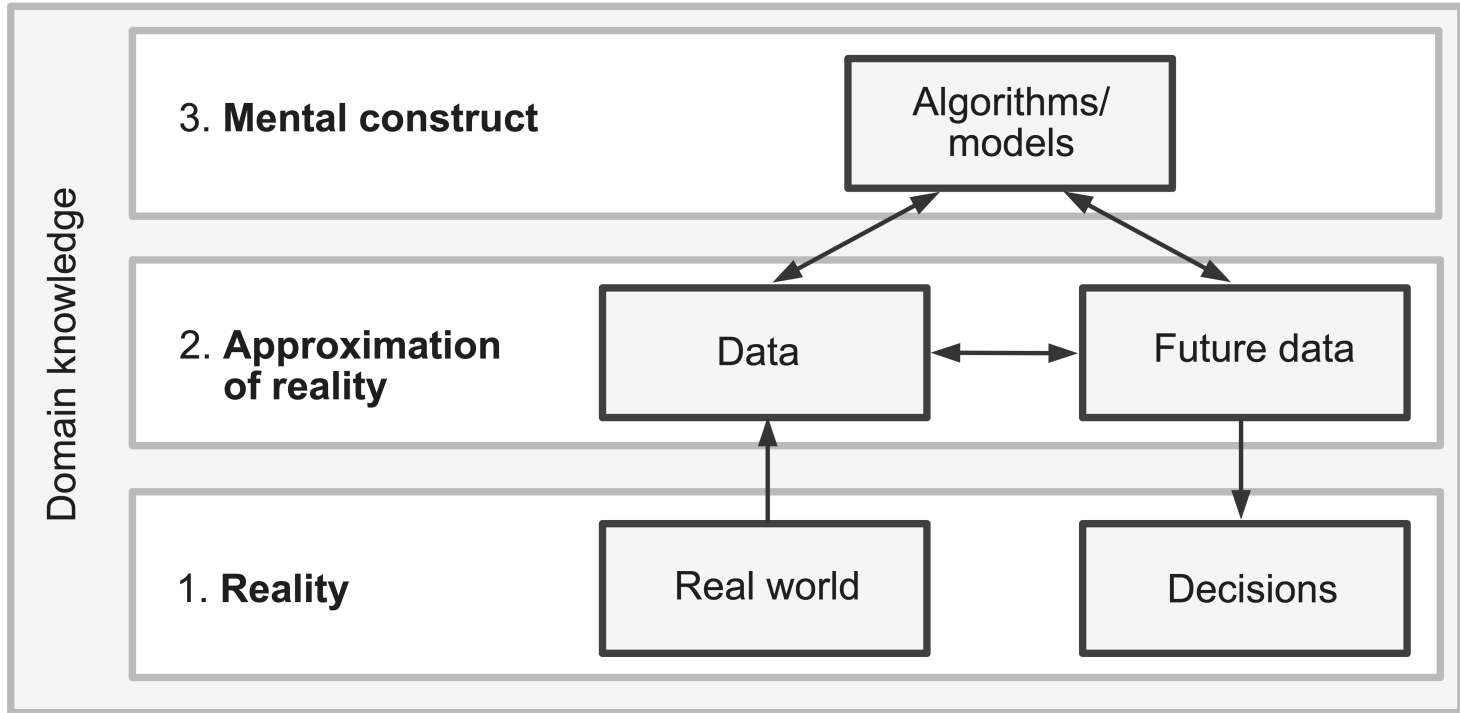


# Prediction

# Using logistic regression



Is our model good at predicting the response?

# Making predictions with the Titanic data

- + For each passenger, we calculate  $\hat{\pi}_i$  (estimated probability of survival)
- + But, we want to predict *which* passengers actually survive

How do we turn  $\hat{\pi}_i$  into a binary prediction of survival / no survival?

$$\begin{array}{lcl} \hat{\pi}_i & \text{predicted probability} & \\ \hat{y}_i & = & \begin{cases} 1 \\ 0 \end{cases} \\ \uparrow & & \\ \text{predicted outcome} & & \end{array} \quad \begin{array}{l} \hat{\pi}_i \geq \text{threshold} \\ \hat{\pi}_i < \text{threshold} \end{array}$$

common threshold : 0.5

# Confusion matrix

```
m1 <- glm(Survived ~ Age + Sex + as.factor(Pclass),  
          data = titanic, family = binomial)
```

```
table(Prediction = m1$fitted.values > 0.5,  
      Truth = titanic$Survived)
```

|                  |       | Truth $\swarrow \begin{matrix} \gamma=0 \\ \gamma=1 \end{matrix}$ |     |
|------------------|-------|-------------------------------------------------------------------|-----|
| Prediction       |       | 0                                                                 | 1   |
| $\hat{\gamma}=0$ | FALSE | 356                                                               | 83  |
| $\hat{\gamma}=1$ | TRUE  | 68                                                                | 207 |

356 = # passengers  $\hat{\gamma}=0, \gamma=0$   
207 = #  $\hat{\gamma}=1, \gamma=1$

Did we do a good job predicting survival?

# Confusion matrix

| ##            | Truth |     |  |
|---------------|-------|-----|--|
| ## Prediction | 0     | 1   |  |
| ## FALSE      | 356   | 83  |  |
| ## TRUE       | 68    | 207 |  |

|             | $Y=0$               | $Y=1$               |
|-------------|---------------------|---------------------|
| $\hat{Y}=0$ | True negative (TN)  | False negative (FN) |
| $\hat{Y}=1$ | False positive (FP) | True positive (TP)  |

Accuracy :  $\frac{\# \text{ correct predictions}}{\# \text{ observations}} = \frac{TP + TN}{n} = \frac{356 + 207}{714}$

$\approx 0.79$

Sensitivity :  $P(\hat{Y}=1 | Y=1) = \frac{TP}{TP + FN} = \frac{207}{207 + 83} = 0.71$   
(TPR)

Specificity :  $P(\hat{Y}=0 | Y=0) = \frac{TN}{TN + FP} = \frac{356}{356 + 68} = 0.84$   
(TNR)

Positive predictive value :  $P(Y=1 | \hat{Y}=1) = \frac{TP}{TP + FP} = \frac{207}{207 + 68} = 0.75$   
(PPV)

## Class activity

- + Work on the class activity (course website)
- + We will then discuss as a group
- + At the end of class, submit your rendered HTML on Canvas

default=1 → business defaulted  
(did not pay back loan)

## Class activity

```
m1 <- glm(Default ~ log(DisbursementGross) + Term +
           as.factor(UrbanRural),
           family = binomial, data = sba)
```

```
table(Prediction = m1$fitted.values > 0.5,
      Truth = sba$Default)
```

##                      Truth  $\swarrow \gamma=0$   $\nwarrow \gamma=1$   
 ## Prediction FALSE TRUE  
 ##  $\hat{\gamma}=0$  FALSE 3989 734  
 ##  $\hat{\gamma}=1$  TRUE 100 168  
 Accuracy = sensitivity  $\cdot P(\gamma=1)$  + specificity  $\cdot P(\gamma=0)$   
 = 0.833  
 + Accuracy =  $\frac{168 + 3989}{3989 + 734 + 100 + 168} = 0.833$   
 + Sensitivity =  $\frac{168}{168 + 734} = 0.19$   
 + Specificity =  $\frac{3989}{3989 + 100} = 0.975$   
 + PPV =  $\frac{168}{100 + 168} = 0.63$

## Class activity

```
##                Truth
## Prediction FALSE TRUE
##      FALSE  3989  734
##      TRUE   100  168
```

Is an accuracy of around 80% good?

## Class activity

Changing thresholds:

As threshold  $\uparrow$ , sensitivity  $\downarrow$   
specificity  $\uparrow$   
As threshold  $\downarrow$ , sensitivity  $\uparrow$   
specificity  $\downarrow$

```
table(Prediction = m1$fitted.values > 0.3,  
      Truth = sba$Default)
```

```
##           Truth  
## Prediction FALSE TRUE  
##      FALSE  3524  351  
##      TRUE   565  551
```

Could choose threshold to  
optimize combo of  
sens & spec.

•  $\text{sens.} + \text{spec}$   
•  $\sqrt{\text{spec} \cdot \text{sens}}$

```
table(Prediction = m1$fitted.values > 0.7,  
      Truth = sba$Default)
```

```
##           Truth  
## Prediction FALSE TRUE  
##      FALSE  4089  902
```